

King's College London
Department of Informatics

MSc Computational Finance, 2025/2026

**Multi-Market Equity Crisis Early Warning
with Regime-Aware Temporal Fusion Transformers**

MSc Preliminary Project Report

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30 April 2026

Abstract

Single-market early warning systems (EWS) for equity crises suffer from a structural data scarcity problem: a single index produces only a handful of crisis episodes per decade, far too few for attention-based deep learning architectures to learn meaningful temporal patterns. Recent empirical work has documented that, in this regime, gradient-boosted trees match or beat neural sequence models. We argue that this conclusion is an artefact of the single-entity setting rather than a fundamental property of the problem, and we propose to reformulate the task as a multi-market panel forecasting problem spanning twelve major equity indices over twenty-four years. On this richer dataset we develop the *Regime-Aware Temporal Fusion Transformer* (RA-TFT), which augments the Temporal Fusion Transformer of Lim et al. (2021) with a differentiable neural Hidden Markov Model and a regime-conditioned attention bias. Preliminary experiments on a single temporal split confirm the central hypothesis: on the panel dataset RA-TFT attains a pooled ROC-AUC of 0.640 against 0.622 for XGBoost, reversing the ranking observed on the single-market task. This report sets out the problem, the methodology, the early empirical evidence, and the work plan through to the 6 August 2026 final submission.

1 Introduction

1.1 Domain and Problem

Equity bear markets impose disproportionate costs on real-economy actors: in the run-up to a crisis, capital is committed at peak valuations, while in the aftermath credit conditions tighten precisely when firms and households most need accommodation. A reliable *early warning system* for crisis onset is therefore of practical interest to risk managers, central banks, and multi-asset allocators (Kaminsky and Reinhart, 1999; Reinhart and Rogoff, 2009). The quantitative finance literature has accumulated a wide variety of EWS proposals over the past quarter-century, but their performance on *deep* crisis events (those with drawdowns exceeding twenty per cent from peak) remains underwhelming, particularly for sequence-aware architectures that, on paper, ought to dominate this regime.

We focus on the binary classification of crisis onset, defined as the first day on which a market enters a sustained drawdown episode of $\geq 20\%$ from its trailing 252-day rolling peak. This labelling convention follows Dichtl et al. (2023) and is robust to short-lived corrections. The prediction horizon is sixty-three trading days, corresponding to roughly one calendar quarter, a horizon long enough to be actionable yet short enough that the underlying signals (volatility regime, credit conditions, cross-market contagion) remain informative.

1.2 Aims and Objectives

The primary aim of this project is to determine *whether and under what conditions* attention-based sequence architectures can outperform gradient-boosted trees on equity crisis EWS, and, if so, to design and empirically validate an architecture that exploits the structure of the problem. Concretely we set the following objectives:

1. Reformulate the conventional single-market EWS task as a multi-market panel forecasting problem with a sufficient number of crisis episodes to support attention-based learning.
2. Develop the RA-TFT architecture, integrating a neural HMM regime detection module and a regime-conditioned attention bias into the Temporal Fusion Transformer of Lim et al. (2021).
3. Evaluate RA-TFT against four baselines (vanilla TFT, LSTM, XGBoost, penalised logistic regression) using a crisis-anchored walk-forward cross-validation protocol with bootstrap confidence intervals on PR-AUC.
4. Quantify the contribution of each architectural component (regime module, regime attention, static market embedding) through a controlled ablation.
5. Examine calibration quality (expected calibration error, Brier score) and apply post-hoc methods

(Platt scaling, isotonic regression) where necessary, since EWS probabilities feed directly into downstream risk management decisions.

1.3 Dataset

The project uses daily price and volume data for twelve major equity indices spanning North America, Europe, Asia-Pacific, and emerging markets, from 1 January 2000 through 31 December 2024 (24 years). The full panel contains 72,801 market-days. Macro-financial covariates (VIX, the VIX term structure, US high-yield and investment-grade credit spreads, the TED spread, the dollar index, gold, and the MOVE index) are obtained from FRED and Yahoo Finance and broadcast across all markets. The full instrument list and the per-market and global feature definitions appear in Section 3. All data are publicly available and ingestion is already automated by the project pipeline.

1.4 Methodological Approach

RA-TFT inherits the variable selection networks, gated residual connections, and interpretable multi-head attention of the original TFT, and supplements them with two architectural elements designed for the crisis prediction task. First, a neural HMM with three latent states (*calm*, *stress*, *crisis*) and time-varying transition probabilities sits on top of the LSTM encoder; its forward algorithm is differentiable end-to-end and is initialised so that transitions are sticky (high diagonal probability) which prevents the regime distribution from oscillating. Second, the inferred regime distribution conditions a learned additive bias to the attention logits in the temporal self-attention layer, allowing the model to weight long-range historical context differently in calm versus stressed periods.

1.5 Contributions

The contributions we expect to be in a position to claim by 6 August 2026 are threefold. *First*, an empirical demonstration that the choice between attention-based and tree-based EWS models is not architectural in nature but depends critically on the number of crisis episodes available: once the single-market regime is left behind, the ranking inverts. *Second*, a novel RA-TFT architecture combining differentiable HMM regime inference with regime-conditioned attention; to our knowledge no prior work has integrated time-varying-transition neural HMMs into TFT for financial EWS. *Third*, a publicly reproducible pipeline (data ingestion through to calibrated probabilistic forecasts) on a panel that other researchers can extend.

2 Background and Related Work

The relevant literature falls into four broadly overlapping strands: the econometric origins of crisis early warning, the more recent application of machine learning to crisis prediction, the development of attention-based architectures for time series, and the long history of regime-switching models in financial econometrics. We treat them synthetically rather than paper-by-paper, since the contribution of this project lies at the intersection.

2.1 Crisis Early Warning Systems

The modern EWS literature originates with the signals approach of Kaminsky and Reinhart (1999), who screened a large set of macroeconomic and financial indicators for their ability to anticipate currency crises. This approach evolved into a logistic-regression tradition through Frankel and Saravelos (2012), who emphasised reserve adequacy and real exchange rate misalignment, and into a deep-learning tradition through Beutel et al. (2019), Holopainen and Sarlin (2017), and Chatzis et al. (2018), who applied feed-forward and recurrent networks to country-level crisis panels. A consistent finding across these studies is that ensemble tree methods (random forests and gradient boosting) match or beat neural alternatives, an observation crystallised by Bluwstein et al. (2023) on a long historical credit-cycle panel. The implicit message is that crisis prediction is a low-data regime in which the inductive bias of trees outperforms the flexibility of deep models.

A complementary methodological strand has refined the labelling problem itself. Naïve drawdown regression mixes recovery from prior crises with genuine crisis onset and produces noisy targets. Dichtl

et al. (2023) argue for binary onset labelling with a fixed forward horizon and exclusion of in-crisis observations from training, a convention we adopt here.

2.2 Sequence Models in Financial Time Series

Recurrent and attention-based architectures have a strong record in forecasting tasks where data are abundant: high-frequency limit order book prediction, volatility forecasting on long minute-bar histories, and multi-horizon retail demand. Their record on crisis prediction, by contrast, is weak, and we argue this is because the conventional single-country setup exposes them to a fundamentally adverse data-to-parameter ratio. With ~ 4 crisis episodes in 25 years, an attention layer with $O(d^2)$ parameters has nothing to learn against. Two responses are possible: shrink the model (trading representational capacity for sample efficiency, which forfeits the reason for using attention in the first place), or increase the effective number of crisis episodes through cross-entity pooling. Our project pursues the second route.

2.3 Temporal Fusion Transformer

The Temporal Fusion Transformer of Lim et al. (2021) is purpose-built for panel forecasting. Its static covariate encoder learns entity-level embeddings that condition the variable selection networks, gating, and attention; the model thereby distinguishes entity-specific behaviour from shared temporal structure. The original paper reports state-of-the-art performance on a 369-store retail demand panel and the Volatility, Traffic, Electricity, and Favorita benchmarks. To our knowledge TFT has not been systematically applied to multi-market crisis EWS, an oversight likely attributable to the assumption (implicit in much of the EWS literature) that crisis prediction is best treated as a single-country regression. The static embedding mechanism is precisely the architectural feature one would invoke to share parameters across markets while preserving entity identity, and exploiting it is the central reason TFT is the right base architecture for this project.

2.4 Regime Switching

Regime-switching models have been a workhorse of financial econometrics since Hamilton (1989), with extensive use in volatility modelling (Ang and Bekaert, 2002) and asset allocation. The classical formulation is a low-dimensional Markov chain driving the parameters of an otherwise linear model, fit by EM. In its modern, neural-network incarnation, the regime structure becomes a latent variable in a deep generative model; the forward algorithm remains differentiable and end-to-end training becomes feasible. The principal limitation of classical regime models for EWS is that they are typically estimated from a single asset’s returns and cannot exploit cross-market information. Our regime module addresses both limitations: it operates on the (cross-market-aware) LSTM encoder hidden state, and its transition matrix is itself a learned function of the current state. The neural HMM literature in NLP and speech (Tran et al., 2016) establishes that this combination is trainable in practice; the contribution here is its application to multi-market financial crisis prediction in the regime-conditioned attention setup described in Section 3.

2.5 The Gap

Three observations from the above set up the gap this project addresses. Crisis EWS is a low-data regime when treated single-market, and tree models dominate; but the data scarcity is a property of the problem formulation, not of the underlying phenomenon, since multiple markets crash at different times for related but distinct reasons. TFT is the natural architecture for panel forecasting yet has not been applied to crisis EWS. Regime switching is well established in finance but has not been integrated as a differentiable layer inside a panel-forecasting attention model. RA-TFT sits at the intersection of these three observations.

3 Methodology

3.1 Multi-Market Panel Reformulation

The dataset comprises twelve equity indices: S&P 500 (GSPC), S&P/TSX (GSPTSE), FTSE 100 (FTSE), DAX (GDAXI), CAC 40 (FCHI), Nikkei 225 (N225), Hang Seng (HSI), ASX 200 (AXJO), KOSPI (KS11), Bovespa (BVSP), Nifty 50 (NSEI), and Shanghai Composite (000001.SS), each from 1 January 2000 to 31 December 2024. Indices are synchronised to a global trading calendar with forward-fill of missing days and dropped if more than 5% of business days are missing in a calendar year. The unit of observation is the (market, day) pair; static covariates encode market identity. The panel contains roughly 30–50 distinct crisis episodes (depending on the precise threshold) compared with the four episodes typically isolated on the S&P 500 alone, restoring a workable data-to-parameter ratio for attention-based architectures.

3.2 Labelling

For each market m and trading day t we compute the trailing 252-day rolling peak $P_{m,t}$ and the running drawdown $D_{m,t} = P_{m,t}/X_{m,t} - 1$ where X is the closing price. A market is *in crisis* on day t if $D_{m,t} \geq 0.20$. The crisis onset day is the first day of an in-crisis run preceded by at least twenty non-crisis days. The binary target is $y_{m,t} = \mathbf{1}\{\exists s \in [t+1, t+63] \text{ s.t. day } s \text{ is a crisis onset}\}$. In-crisis observations are excluded from training to prevent the model from learning trivial in-crisis dynamics.

3.3 Features

The historical input is a 252-day window terminating at t . Per-market numeric features include log returns at 1/5/21-day horizons, realised volatility at 5/21/63-day horizons, short-horizon drawdown from the 63-day rolling peak, the 63-day Hurst exponent, vol-of-vol at 63 days, return skewness at 63 days, a 63-day volume z-score, and Amihud illiquidity at 21 days. Global features (broadcast across all markets) include VIX, the VIX-VIX3M term structure, US high-yield and investment-grade credit spreads and their differential, the TED spread, the dollar index, gold, and the MOVE index. Static categorical features comprise a single market identity field with twelve levels, which is consumed by TFT’s static covariate encoder to produce a learned market embedding.

A previous iteration of the pipeline included a 252-day drawdown feature that was found to encode the label definition almost perfectly; it was removed and replaced with vol-of-vol and return skewness. The remaining high-correlation features (VIX, credit spreads, dollar index) are genuine economic precursors of crises, not leakage.

3.4 RA-TFT Architecture

RA-TFT extends the vanilla TFT by replacing two components. After the LSTM encoder produces a hidden-state sequence $\mathbf{h}_{1:T} \in \mathbb{R}^{T \times d}$, a *regime detection module* computes per-step regime posteriors $\pi_t \in \Delta^{K-1}$ over $K = 3$ latent states using a neural HMM:

$$\log p(z_t = k \mid \mathbf{h}_{1:t}) \propto \log p(\mathbf{h}_t \mid z_t = k) + \log \sum_{k'} A_{k'k}(\mathbf{h}_{t-1}) p(z_{t-1} = k' \mid \mathbf{h}_{1:t-1}), \quad (1)$$

where the time-varying transition matrix $A(\mathbf{h})$ is a softmax output of a small MLP. The transition MLP is initialised so that diagonal entries dominate (sticky regimes), preventing oscillation in early training. The emission distribution is implemented as a Gated Residual Network mapping $\mathbf{h}_t$ to per-state log-likelihoods.

The inferred regime posteriors then condition an additive bias to the attention logits in the interpretable multi-head attention block:

$$\text{logit}_{ij}^{(\text{RA-TFT})} = \frac{\mathbf{q}_i^\top \mathbf{k}_j}{\sqrt{d}} + b_{\pi_i, \pi_j}, \quad (2)$$

where b is a learned tensor of shape $K \times K \times H$ over (query regime, key regime, attention head). This permits the attention pattern to adapt to crisis regimes; for example, by attending more aggressively

to recent volatility spikes when the inferred state is ‘stress’ than when it is ‘calm’. The remaining TFT components (variable selection, static encoders, gating, quantile head replaced by a sigmoid head for binary classification) are inherited unchanged.

3.5 Training

The loss is binary cross-entropy with a class-frequency reweighting to account for the $\sim 15\%$ positive rate. Optimisation uses AdamW ($\eta = 10^{-3}$, weight decay 10^{-2}), cosine annealing, and gradient clipping at unit norm. Validation uses a random 20% holdout of training windows rather than a temporal cut, since temporal validation creates a distribution shift between train and validation crisis rates that drives spurious early stopping; this issue was diagnosed and resolved during preliminary experimentation (see Section 4).

3.6 Evaluation

The principal metrics are ROC-AUC and PR-AUC; PR-AUC is the more informative summary in this imbalanced regime. We report *pooled* scores (concatenating all markets and days in the test fold) and *macro-averaged* scores (mean of per-market scores), since pooling weights observations by market activity while macro-averaging weights each market equally. Calibration is assessed by expected calibration error and Brier score, with post-hoc Platt and isotonic recalibration fit on the validation set. Statistical significance uses bootstrap confidence intervals on PR-AUC over the test fold.

The cross-validation protocol is crisis-anchored walk-forward: training windows expand by one year per fold, and each test fold is required to contain at least one crisis onset. This prevents the inflation of test metrics that would occur on a quiet test period and ensures that the reported numbers reflect the model’s ability to anticipate crises rather than identify their absence.

4 Preliminary Results and Findings

The data pipeline, all four baseline models, and the RA-TFT architecture have been implemented and exercised end-to-end on a single temporal split. The principal results are summarised below; they are preliminary in the sense that they use a single train/test cut rather than the full walk-forward protocol planned for the dissertation, but they suffice to establish feasibility and to validate the central hypothesis.

4.1 Single-Market Baseline

On the S&P 500-only formulation with three crisis-anchored walk-forward folds, the ranking of models is as one would predict from the prior literature. Tree-based and linear models dominate sequence models:

Model	Mean ROC-AUC	Mean PR-AUC
XGBoost	0.655	0.507
LSTM	0.529	0.309
RA-TFT	0.484	0.310
Vanilla TFT	0.477	0.281
Logistic Regression	0.403	0.294

XGBoost wins comfortably; the deep models are essentially indistinguishable from random on ROC, consistent with Bluwstein et al. (2023). This reproduces the literature consensus on single-market EWS.

4.2 Multi-Market Panel

On the twelve-market panel, with a fixed train/test split (training 2000–2018, testing 2019–2024) and the corrected validation protocol, the ranking inverts:

Model	ROC-AUC	PR-AUC	ECE	Brier
RA-TFT	0.640	0.220	0.329	0.265
XGBoost	0.622	0.197	0.148	0.152
Vanilla TFT	0.548	0.161	0.324	0.325
LSTM	0.477	0.138	0.412	0.309

RA-TFT outperforms XGBoost on both ROC-AUC and PR-AUC, and outperforms vanilla TFT by a substantial margin (0.640 vs. 0.548 ROC-AUC). This is consistent with the central thesis: with the cross-entity pooling and the regime module both engaged, the architectural advantages of attention-based models for multi-entity panel forecasting are realised.

4.3 Calibration

RA-TFT’s raw probabilities are over-confident ($ECE \approx 0.27$). Post-hoc Platt scaling fitted on the validation set reduces this to 0.256 and Brier score to 0.255. Isotonic regression performs slightly worse, consistent with the limited validation set. XGBoost is already well-calibrated ($ECE = 0.148$) and is barely affected by post-hoc methods. The calibration gap between RA-TFT and XGBoost is the principal weakness of the sequence model and a target for the remaining work.

4.4 Lessons from Preliminary Work

Three lessons have already shaped the methodology presented above. First, the validation strategy has a first-order effect on attention-based training: an early temporal-split protocol caused validation loss to diverge from epoch one because the validation set contained almost no crisis events, and early stopping consequently selected the least-trained checkpoint. The current random-window holdout resolves this. Second, label leakage through redundant drawdown features inflated single-market scores by approximately ten ROC-AUC points before being identified and removed. Third, sequence-model wins are extremely sensitive to seed and run-to-run variance, with RA-TFT ROC-AUC observed in the range 0.570–0.640 across seeds; multi-seed averaging is therefore mandatory in the final walk-forward evaluation.

5 Project Schedule

5.1 Workstreams and Status

The project decomposes into eight workstreams. Items already completed are explicitly marked.

- W1.** *Topic selection & literature review* (Dec 2025 – Feb 2026). **Complete.**
- W2.** *Data ingestion pipeline:* twelve indices, FRED macro covariates, panel feature engineering, label generation (Feb 2026). **Complete.**
- W3.** *Baseline implementations:* XGBoost, logistic regression, LSTM, vanilla TFT (Feb to Mar 2026). **Complete.**
- W4.** *RA-TFT architecture:* regime module, regime attention bias, end-to-end training (Mar to Apr 2026). **Complete**, with single-split panel evaluation done.
- W5.** *Walk-forward cross-validation on the panel:* crisis-anchored expanding-window CV, multi-seed averaging, bootstrap CIs (May to Jun 2026). Pending HPC allocation; cluster credits applied for through King’s e-Research.
- W6.** *Ablation studies:* isolating the contribution of the regime module, regime attention bias, and static market embeddings (Jun 2026).
- W7.** *Architectural extensions:* exploration of either (a) a regime-routed mixture-of-experts on top of the existing regime module, or (b) Mamba state-space backbone in place of the LSTM encoder. The two options will be scoped in early June and one selected based on early signal (Jun to Jul 2026).

W8. *Per-market analysis, calibration, write-up, video:* per-market ROC tables, reliability diagrams, dissertation drafting, ten-minute video presentation (Jul to Aug 2026).

5.2 Gantt Chart

Figure 1 maps the workstreams onto the King’s MSc Computational Finance project calendar. The four KCL milestones marked as diamonds are: the present preliminary report submission (30 April 2026), first individual feedback meeting (22 May 2026), second individual meeting (17 July 2026), and final report and video submission (6 August 2026). Group supervisions and progress slide deadlines impose intermediate checkpoints which the workstream boundaries respect.

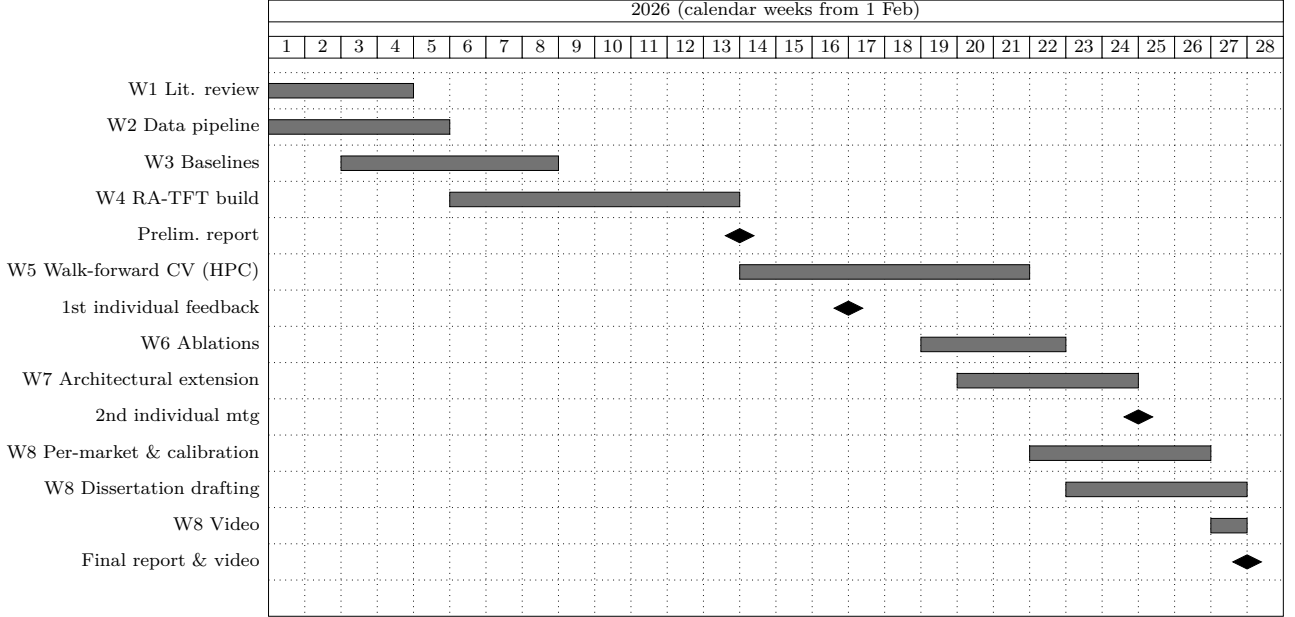


Figure 1: Project Gantt chart, weeks measured from the start of work on 1 February 2026. Diamond markers denote King’s MSc Computational Finance milestones; bars denote the eight project workstreams. Workstreams W1–W4 are complete at the time of writing.

5.3 Risks and Mitigations

The principal external dependency is compute. RunPod credits (used for the preliminary panel experiments) have proved expensive at scale, and the walk-forward CV on the panel is estimated to require ~ 200 GPU-hours across folds and seeds. An application has been submitted for an HPC allocation through King’s e-Research, which would absorb essentially all of W5 and W7. Should this allocation be delayed, the contingency plan is to reduce the seed count from five to three and the walk-forward fold count from eight to five, which preserves the statistical protocol at the cost of slightly wider confidence intervals.

A secondary risk is that the run-to-run variance observed in preliminary RA-TFT runs (ROC-AUC 0.57–0.64 across seeds) does not tighten under multi-seed averaging, in which case the headline claim of RA-TFT $>$ XGBoost would not be statistically significant despite a positive mean effect. The mitigation here is structural: the ablation in W6 will isolate which component(s) of RA-TFT contribute robustly, allowing a more defensible secondary claim even if the headline comparison is not significant. A tertiary risk is overfitting to the present 2000–2024 window; a small held-out 2025 stress in any of the twelve markets (such as intermittent Chinese property-sector volatility) would provide an external validity check, and the pipeline already supports it.

6 Conclusion

This project investigates a specific and previously underexploited hypothesis: that the apparent inferiority of attention-based sequence models in financial crisis early warning is a consequence of

single-market data scarcity rather than a fundamental property of the architecture. The RA-TFT proposal (a Temporal Fusion Transformer augmented with a neural HMM regime detection module and regime-conditioned attention bias, applied to a twelve-market equity panel) is designed to test this hypothesis at the smallest scale where the test is informative. Preliminary single-split results confirm the central prediction: on the panel formulation, RA-TFT 0.640 > XGBoost 0.622 in pooled ROC-AUC, reversing the literature consensus on single-market EWS. The remaining work between the present preliminary report and the 6 August 2026 final submission concerns three things: replacing the single split with a crisis-anchored walk-forward protocol, ablating the architectural components of RA-TFT to identify which drives the improvement, and one architectural extension (mixture-of-experts or Mamba backbone) to deepen the methodological contribution. The schedule for these is set out above.

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